



BITS Pilani
DIGITAL

READER NOTES

FUNDAMENTALS OF FINANCE | MODULE 1

Understanding Financial Markets



STUDY NOTES • EDITION 1.0 • WEEK 1

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How to use these notes



These notes cover the whole of Week 1: every concept and every worked example, written for reading and recall rather than for listening. They are yours to keep, mark up and revise from.

The week has nine topics in three parts. Part A defines what finance is and where it happens. Part B covers how financial decisions are made and the principles behind them. Part C looks at who participates and how the machinery actually works. Read the parts in order the first time; each assumes the last.

When assessment approaches you will not have time to re-read everything, and you will not need to. The Quick Recap boxes, the module summary and the self-check at the back were written for exactly that week. A workable plan: recaps first, then the summary, then attempt the self-check from memory before looking at the answers.

What each topic contains

Every topic follows the same rhythm, so you always know where to look.

Section	What it gives you
In one line	The whole idea compressed to a single sentence
Why this matters	Where the concept sits in the course, and why it earns your attention
The core idea	The full explanation, developed step by step
Figure	A diagram or comparison that makes the idea visible
Callouts	Key ideas, worked examples, pitfalls and exam pointers
Quick recap	Five bullets to revise from

Callouts used in this document

KEY IDEA

The most important takeaway of a section. If you remember nothing else, remember this.

COMMON PITFALL

A mistake learners make repeatedly, in assignments and in examinations.

EXAM POINTER

How the concept is typically tested, and what a full answer must contain.

REAL-WORLD EXAMPLE

A concrete case showing the concept at work outside the classroom.

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3	Finance, economics and the three facets

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C Who participates, and how the machinery works

7	The four types of investors
8	Flow of funds and financial intermediaries
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CONSOLIDATION

Module summary, by learning objective

Glossary and self-check

Module overview



This module introduces finance as a field: what it is, the environment in which it operates, the principles that govern every financial decision, and the markets, participants and mechanisms through which capital actually moves. It closes on price discovery, the process that turns information into the prices which allocate capital across an economy.

One thread runs through all nine topics: **finance exists to connect savers with borrowers**. Every institution, market and instrument in this course is a mechanism for closing that gap efficiently, safely and at a fair price. Hold on to that and the rest of the programme has a spine.

Learning objectives

By the end of this module, you will be able to:

- 1 **Define** finance, name the three core financial decisions, and describe the six components and four nested layers of the financial environment.
- 2 **Explain** the six principles of finance and apply the time-value-of-money formula to a numerical example.
- 3 **Classify** financial markets by maturity, newness and structure, and distinguish the four types of investors.
- 4 **Explain** how funds flow from savers to borrowers, why intermediaries exist, and how price discovery allocates capital.

Topic map

9 topics, three parts. The Source column gives the lecture video each topic draws on, so you can go back to the original; the Serves column shows which objective it supports.

Topic	What it covers	Part	Source	Serves
1	What finance is, and why it matters	A • Foundations	L1 V1	LO 1
2	The financial environment and its nested layers	A • Foundations	L1 V1	LO 1
3	Finance, economics and the three facets	A • Foundations	L1 V2	LO 1
4	How financial decisions get made	B • Principles	L2 V1	LO 2
5	The six principles of finance	B • Principles	L2 V2	LO 2
6	The financial system and its markets	B • Principles	L3 V1	LO 3
7	The four types of investors	C • Mechanics	L3 V2	LO 3
8	Flow of funds and financial intermediaries	C • Mechanics	L4 V1	LO 4
9	Price discovery	C • Mechanics	L4 V2	LO 4

Key terms at a glance



Fuller definitions appear in the glossary at the back. Skim these before your first read; they carry most of the week's vocabulary.

Finance

How individuals, institutions, governments and businesses acquire, spend and manage money and other financial assets.

Nested model

Environment → institutions → markets → investments. Each layer depends on the one outside it.

Patient capital

Funding willing to wait for returns over a long horizon, supporting innovation and growth.

Diversification

Spreading investments across assets to cut risk without proportionally cutting expected return.

Principal-agent problem

The conflict when managers (agents) pursue goals different from shareholders (principals).

Capital market

Instruments maturing beyond one year: bonds, mortgages and equity.

Secondary market

Where existing securities trade between investors; supplies liquidity and price signals.

Direct finance

Borrowers raise funds straight from lenders by issuing instruments through markets.

Asset transformation

Borrowing short and lending long, converting illiquid risky assets into safer liquid ones.

Moral hazard

After the loan: borrowers take greater risks once funded. Addressed by monitoring.

Financial environment

The six interdependent components within which all financial activity happens.

NBFC

Non-banking finance company. Provides financial services without a full banking licence.

Time value of money

A rupee today is worth more than a rupee tomorrow, because it can be invested.

Market efficiency

The condition in which prices already reflect all available public information.

Money market

Instruments maturing in one year or less. High liquidity, low risk, modest return.

Primary market

Where securities are issued for the first time; proceeds go to the issuer.

Institutional investor

An entity investing other people's money — the defining feature, not size.

Indirect finance

Funds routed to borrowers through an intermediary. Most global financing works this way.

Adverse selection

Before the loan: the riskiest borrowers are keenest to borrow. Addressed by screening.

Price discovery

The process by which buyer and seller interaction determines a security's price.

PART A • THE FOUNDATIONS

What finance is, and where it happens

1 What Finance Is, and Why It Matters

PART A • THE FOUNDATIONS • TOPIC 1 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

You have been making spending, saving and investing decisions your whole life. Finance is the name for them, and the study of how they scale from a household to an economy.

Why this matters

Everything in this course sits on one definition and three decisions. Get them precise now and the rest of the week is elaboration; carry a vague version forward and the vocabulary of Weeks 2 to 4 will not attach to anything.

A working definition

It is the first of the month and your salary of ₹45,000 has arrived. Within thirty seconds, before you open your banking app, you will make three decisions: how much to spend, how much to set aside, and where to put what is left.

Melicher and Norton define finance as *the study of how individuals, institutions, governments and businesses acquire, spend and manage money and other financial assets*. Note how wide that is. It covers a household choosing between a fixed deposit and a systematic investment plan, a company deciding whether to fund a plant with debt or equity, and a government financing a highway. Different scales, one question: where should this money go, and why there rather than somewhere else?

Why it matters at three levels

For individuals, finance shapes personal wealth. Two people on identical salaries can reach very different positions in their fifties purely through the quality of decisions made in their thirties. **For businesses**, it drives expansion, hiring and market entry — and poor financial judgement destroys firms whose products are excellent. **For the economy**, efficient fund flows let firms invest, create jobs and raise living standards.

The three core financial decisions

Every actor — individual, business or government — faces the same three. **Spending** allocates funds today, on consumption, operations or capital expenditure. **Saving** sets funds aside for future needs. **Investing** deploys funds into assets expected to grow or generate returns. These three form the foundation of financial behaviour at every level of the system.

THE THREE DECISIONS EVERY FINANCIAL ACTOR MAKES

SPENDING

Allocate funds today • consumption, operations, capital expenditure

SAVING

Set aside for future needs • deposits, provident fund

INVESTING

Deploy into assets expected to grow or yield returns

The same three decisions scale from a ₹45,000 salary to a national budget. Only the magnitude changes.

Figure 1.1 The three core financial decisions. Every financial actor faces all three, continuously.

KEY IDEA

Finance is not a specialist activity performed by bankers. It is the name for a decision every person and institution already makes. The discipline exists because those decisions can be made well or badly, and the difference compounds.

EXAM POINTER

“Define finance” is a guaranteed mark. Give the Melicher and Norton wording — acquire, spend and manage money and other financial assets — and name all four actors: individuals, institutions, governments and businesses. Then add the three decisions. An answer that stops at “finance is about money” earns almost nothing.

QUICK RECAP • TOPIC 1

- Finance: how individuals, institutions, governments and businesses acquire, spend and manage money and other financial assets.
- At its simplest, finance is the mechanism connecting savers with borrowers.
- It matters at three levels: personal wealth, business survival, and economic stability.
- Three core decisions run through every level: spending, saving, investing.
- The decisions are universal; only the scale changes.

2 The Financial Environment and Its Nested Layers

PART A • THE FOUNDATIONS • TOPIC 2 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

Six interdependent components make up the financial environment, and they sit inside one another like concentric circles rather than side by side.

Why this matters

Weeks 2 to 4 each take one of these components apart. If you cannot name all six and say how they nest, later material will feel like a list of unrelated institutions instead of one system.

The six components

The financial system is the overarching framework of rules, regulations and infrastructure governing how funds move — in India, the RBI, SEBI and the legislation around them. **Financial institutions** are the intermediaries: banks, non-banking finance companies, insurers. **Financial markets** are the venues, physical and electronic, where instruments trade — the NSE, the bond market. **Businesses** raise capital and deploy it productively. **Individuals** save, borrow, invest and spend, usually through the institutions above. **Global interactions** are the cross-border flows that link India to the rest of the world.

Interdependence is the point

These are not six separate topics. A change in one flows through the rest. When the RBI raises the repo rate, individuals find borrowing more expensive, businesses postpone investment, and markets reprice. The entire environment moves together, which is why the course treats them as one subject.

The nested model

Structurally the components sit inside one another. The **financial environment** is the outermost layer. Within it sit **financial institutions**, the infrastructure through which funds are channelled. Inside institutions sit **financial markets**, the venues through which instruments trade. At the core sit **financial investments** — the shares, bonds, funds and deposits actually held.

Each layer depends on the one outside it. You cannot have efficient investments without functioning markets, nor functioning markets without stable institutions, nor any of it without a sound environment.

FOUR NESTED LAYERS • EACH DEPENDS ON THE ONE OUTSIDE IT

FINANCIAL ENVIRONMENT

System • institutions • markets • businesses • individuals • global interactions

FINANCIAL INSTITUTIONS

Banks, NBFCs, insurers — the channels through which funds move

FINANCIAL MARKETS

The venues where instruments are traded

FINANCIAL INVESTMENTS

Shares • bonds • funds • deposits

Figure 2.1 The nested model. Read it inwards: each layer exists only because the one outside it functions.**REAL-WORLD EXAMPLE**

Priya, twenty-eight, has found a flat priced at ₹60 lakh. She has ₹6 lakh and needs ₹54 lakh. Somewhere in the same city sit people with ₹54 lakh idle. Every one of the six components has to work for those two facts to meet: a bank willing to lend (institution), a rate set within a policy framework (system), a market that prices the money, and a legal structure that makes the promise enforceable.

COMMON PITFALL

Listing the six components as a flat set. The examinable insight is that they *nest* and that a shock to the outer layer propagates inwards. A candidate who writes six bullet points has answered half the question.

QUICK RECAP • TOPIC 2

- Six components: the system, institutions, markets, businesses, individuals, global interactions.
- They are interdependent — an RBI rate change moves all of them together.
- Four nested layers: environment → institutions → markets → investments.
- Each layer depends on the one outside it, never the reverse.
- Investments sit at the core: they are the objects, not the machinery.

3 Finance, Economics and the Three Facets

PART A • THE FOUNDATIONS • TOPIC 3 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

Economics allocates resources; finance allocates the funds that pay for them. And finance itself divides into three overlapping facets that no practitioner works in separately.

Why this matters

Two distinctions that examiners like because they are easy to state and easy to get slightly wrong. Both also set up the structure of the whole programme: markets and institutions is Weeks 1 to 4, investments is Weeks 5 to 10, financial management runs throughout.

The two functional areas

Financial institutions and intermediaries handle the efficient transfer of funds between those who wish to save and those who need to borrow. This includes banks and non-banking finance companies, the latter providing financial services without a full banking licence.

Financial markets are the venues where the exchange of instruments takes place. Individuals rarely operate in them directly — they do so through institutions and platforms such as Zerodha or Groww.

How finance differs from economics

Economics concerns the efficient allocation of *resources* — land, labour, capital — to maximise output or utility, guided by demand and supply toward an equilibrium price. Finance concerns the efficient allocation of *funds* to pay for those resources.

Here is the distinction that carries the marks. In economics the equilibrium price is set by demand and supply. In finance, the correct price must additionally be supported by the **ability to pay**. Priya may agree that ₹60 lakh is the fair market price for that flat; without a lender, the price is academic.

Entrepreneurial and personal finance

Finance looks different depending on who is doing it. **Entrepreneurial finance** covers how firms meet financing needs, and those needs vary by stage. *Patient capital* waits for returns over a long horizon and supports innovation; *quick funding* meets immediate operational needs such as cash flow and supplier payments.

Personal finance covers how individuals manage savings, investment and expenditure across life stages — education, then consumption and housing, then health. This is exactly why products are designed around life stages: HDFC Bank offers housing loans and consumer loans, health insurance covers medical emergencies, and credit cards provide short-term flexible access.

The three facets

Conventionally drawn as three overlapping circles, and the overlap is deliberate. **Markets and institutions** is the architecture of finance. **Investments** is how capital is deployed. **Financial management** is how businesses raise, allocate and steward capital — the CFO's domain. A company raising funds on the NSE engages all three at once.

ECONOMICS ALLOCATES RESOURCES • FINANCE ALLOCATES THE FUNDS THAT PAY FOR THEM

ECONOMICS*"How should scarce resources be allocated?"***ALLOCATES**

Resources — land, labour, capital

OBJECTIVE

Maximise output or utility

PRICE SET BY

Demand and supply equilibrium

FINANCE*"How are those resources paid for?"***ALLOCATES**

Funds — money and financial assets

OBJECTIVE

Pay for resources efficiently

PRICE SET BYEquilibrium price *plus the ability to pay***Figure 3.1** The distinction that matters: in finance, arriving at the right price is not enough.**KEY IDEA**

In economics a price clears the market. In finance a price must also be **funded**. Every instrument you meet for the rest of this course exists to close that gap between a price agreed and a price payable.

EXAM POINTER

"Distinguish finance from economics" should be answered on three axes: what is allocated (resources versus funds), the objective, and how price is determined. The mark that separates a good answer from a full one is the ability-to-pay clause. For the three facets, name all three *and* state that they overlap, with one example of a decision touching all three.

QUICK RECAP • TOPIC 3

- Two functional areas: institutions transfer funds, markets provide the venue.
- Economics allocates resources; finance allocates the funds that pay for them.
- The critical difference: in finance the price must be supported by ability to pay.
- Entrepreneurial finance needs patient capital and quick funding; personal finance follows life stages.
- Three overlapping facets: markets and institutions, investments, financial management.

PART B • PRINCIPLES AND DECISIONS

How decisions are made, and the rules behind them

4 How Financial Decisions Get Made

PART B • PRINCIPLES AND DECISIONS • TOPIC 4 OF 9 • SERVES LEARNING OBJECTIVE 2



IN ONE LINE

Financial decision-making runs on five layers, from policymakers down to the instruments held — and every layer depends on the one above it.

Why this matters

This topic supplies the vocabulary for the whole of Week 2, which takes the monetary system and the banks apart in detail. It also explains why a single policy decision reaches your home loan.

Why it matters at all

On 22 March 2020, as India announced its first national lockdown, a small group at the RBI cut the repo rate by 75 basis points. Within forty-eight hours banks began reducing lending rates; within weeks, borrowing costs fell for crores of people who had never once thought about monetary policy. That is financial decision-making at scale.

It matters for four reasons: it enables informed personal choices; it shapes personal wealth over a lifetime; it supports business judgement about whether a project creates value; and it lets citizens interpret inflation data, rate announcements and budgets.

The five layers

The course follows Melicher and Norton in using the United States system as the worked illustration, so the names are American. The structure is general, and the Indian equivalent is given alongside in the figure. Each layer depends on the one above: investments function because markets operate, markets operate because institutions are stable, institutions operate within a reliable monetary system, and all of it sits inside the framework policymakers set.

Five economic goals, and what shapes a decision

Policymakers typically pursue a balanced budget, lower taxes, free trade, low inflation and full employment.

At the individual level, decisions are shaped by interest rates (borrowing costs and savings returns), inflation (purchasing power), economic conditions (income stability), government policy (taxation, subsidies) and market expectations. Business decisions turn on financing operations, investment choices, cost and profit management, and risk evaluation.

FIVE LAYERS • EACH DEPENDS ON THE ONE ABOVE IT

POLICYMAKERS

US: President, Congress, Treasury, Federal Reserve Board • India: Parliament, Ministry of Finance, RBI, SEBI

MONETARY SYSTEM

Central bank and the commercial banking system • creates and transfers money, controls liquidity

FINANCIAL INSTITUTIONS

Depository institutions, contractual savings organisations, securities firms, finance companies

FINANCIAL MARKETS

Debt • equity • derivatives • foreign exchange

FINANCIAL INVESTMENTS

The instruments actually held

Figure 4.1 The layered structure of a financial system, with the Indian equivalent of each US institution.**REAL-WORLD EXAMPLE**

Trace the March 2020 repo cut down the stack. Policymakers act; the monetary system transmits the change to bank funding costs; institutions reprice their lending; markets reprice bonds and equities; and the investments held by ordinary savers change in value. One decision, five layers, forty-eight hours.

EXAM POINTER

If asked to describe the financial system, name all five layers *in order* and state the dependency direction — each layer rests on the one above. Adding the Indian equivalent of each American institution is an easy way to show command of the material rather than recall of a diagram.

QUICK RECAP • TOPIC 4

- Financial decision-making matters for informed choice, personal wealth, business judgement and civic understanding.
- Five layers: policymakers, monetary system, institutions, markets, investments.
- Each layer depends on the one above it.
- Five economic goals: balanced budget, lower taxes, free trade, low inflation, full employment.
- Personal decisions turn on rates, inflation, conditions, policy and expectations.

5 The Six Principles of Finance

PART B • PRINCIPLES AND DECISIONS • TOPIC 5 OF 9 • SERVES LEARNING OBJECTIVE 2



IN ONE LINE

Six ideas govern every financial decision ever made, from a household savings account to a billion-rupee merger.

Why this matters

This is the most examinable topic in Week 1 and the one that recurs most often later: the time value of money returns in Weeks 5 to 8, risk and reward underpin the whole of portfolio theory, and the principal-agent problem reappears in governance and in Week 2's treatment of investment banks.

A question that needs no calculator

Would you rather have one lakh rupees now, or one lakh rupees exactly one year from today? Almost everyone answers immediately — now, obviously — and almost nobody can explain why. That instinct has a name, and it is the first of six principles forming a framework that applies to individuals, businesses and institutions alike.

The six, in order

1 • Time value of money. Money available today is worth more than the same amount later, because it can be invested to earn returns. $FV = PV \times (1 + r)^t$.

2 • Risk versus reward. Risk is uncertainty about a future outcome. Rational investors accept higher risk only when compensated with the expectation of higher returns. A business investment might return anywhere from nothing to ₹2,40,000; a government security might return between ₹95,000 and ₹1,10,000 — narrow range, lower risk, lower return.

3 • Diversification. Some risk falls simply by spreading investments across assets, because different assets rarely perform poorly at exactly the same time. Diversification shifts likely outcomes towards the middle and away from the extremes.

4 • Market efficiency. Prices reflect all available public information, because informed professionals continually hunt mispricing and correct it by acting on it. Any informational advantage is therefore short-lived.

5 • The principal-agent problem. Shareholders want returns maximised; managers may want higher salaries, power or lower personal risk. Resolved by aligning incentives — performance-linked pay, stock options, ownership stakes.

6 • Trust and ethics. Reputation reflects ethical behaviour over time and is among a firm's most valuable assets. Fraud and misrepresentation destroy it, invite legal action, and in serious cases collapse the business.

THE SIX PRINCIPLES • ONE FRAMEWORK, EVERY LEVEL OF FINANCE

1 • Time value of money

A rupee today beats a rupee tomorrow, because it can earn.

2 • Risk versus reward

Higher risk must carry the expectation of higher return.

3 • Diversification

Spreading assets cuts risk without proportionally cutting return.

4 • Market efficiency

Prices already hold the public information.

5 • Principal-agent

Align manager incentives with owner goals.

6 • Trust and ethics

Reputation is an asset; losing it ends firms.

Figure 5.1 The six principles. Expect to name all six and apply at least one numerically.

KEY IDEA

Compounding is the whole of principle one. ₹1,00,000 at 6% for five years yields **₹33,823**, not ₹30,000. The extra ₹3,823 is interest earned on interest — which is why starting an investment plan at twenty-five rather than thirty-five matters more than most people assume.

EXAM POINTER

Two things are asked here. First, name all six with a one-line explanation of each — partial lists lose marks proportionally. Second, apply the time-value formula and *show the working*: state $FV = PV \times (1 + r)^t$, substitute, then compute. Quoting only the answer forfeits the method marks.

COMMON PITFALL

Treating diversification as “owning many things”. The principle depends on assets not moving together; ten stocks in one sector diversify almost nothing. This distinction is the entire basis of Weeks 5 and 6.

QUICK RECAP • TOPIC 5

- Six principles: time value, risk-reward, diversification, market efficiency, principal-agent, trust and ethics.
- $FV = PV \times (1 + r)^t$. ₹1,00,000 at 6% becomes ₹1,06,000 in one year and ₹1,33,823 in five.
- Rational investors accept higher risk only for higher expected return.
- Diversification works because assets do not move together.
- Aligning manager incentives resolves the principal-agent conflict; trust underpins everything.

6 The Financial System and the Shape of Its Markets

PART B • PRINCIPLES AND DECISIONS • TOPIC 6 OF 9 • SERVES LEARNING OBJECTIVE 3



IN ONE LINE

Markets are classified on three independent dimensions — maturity, newness and structure — and any single security is described by all three at once.

Why this matters

Students routinely confuse “primary versus secondary” with “money versus capital” because the two pairs are introduced in sequence. They are orthogonal. Getting this straight now prevents a recurring error in Weeks 3 and 4.

What the financial system is

A complex mix of intermediaries, markets, instruments, policymakers and regulations that interact to expedite the flow of financial capital from savings into investment. It is a network, not a single institution.

Its three components each perform two functions. The **monetary system** creates and transfers money. **Financial institutions** accumulate savings and lend or invest them. **Financial markets** market financial assets and transfer them between investors.

The institutional hierarchy

At the top sits the central bank, responsible for monetary policy and stability. Below it, payment systems and depository banks. Then non-banking financial companies, providing lending and investment services without a full banking licence. Below them, investment banks, which help companies raise capital and structure deals. At the base, brokerages, executing trades for clients.

Three independent classifications

By maturity. Money markets handle instruments of one year or less — treasury bills, commercial paper, certificates of deposit — with high liquidity, low risk and modest returns. Capital markets handle longer instruments: bonds, mortgages and equity.

By newness. In a primary market, securities are sold for the first time and the proceeds go to the issuer. In a secondary market, existing securities change hands between investors and the issuer receives nothing — but liquidity, risk reallocation and price signals are generated.

By structure. Organised exchanges are centralised and standardised, open to retail participation. Over-the-counter markets are decentralised dealer networks, used largely by institutions.

THREE INDEPENDENT DIMENSIONS • EVERY SECURITY IS DESCRIBED BY ALL THREE
BY MATURITY
MONEY MARKET

One year or less • T-bills, commercial paper, CDs • high liquidity, low risk

CAPITAL MARKET

Over one year • bonds, mortgages, equity • higher risk and return

BY NEWNESS
PRIMARY

First issue • proceeds go to the issuer

SECONDARY

Existing securities • proceeds go to the seller • supplies liquidity and prices

BY STRUCTURE
EXCHANGE

Centralised, standardised, retail access

OVER-THE-COUNTER

Decentralised dealer networks, largely institutional

*A newly issued government bond traded on an exchange is **capital market** • **primary** • **exchange** — all three at once.*

Figure 6.1 The three classifications are orthogonal. Any security carries one label from each column.

COMMON PITFALL

Treating the three dimensions as a single sequence, so that “secondary” is somehow opposed to “capital market”. They are independent axes. A share traded on the NSE today is capital market, secondary and exchange simultaneously.

EXAM POINTER

Classification questions are common and easy to score fully. Define both members of each pair, state what distinguishes them, and then give one security labelled on all three axes. If the question mentions the eight market categories, name them: debt securities, debt, bond, mortgage, equity securities, equity, derivative and foreign exchange markets.

QUICK RECAP • TOPIC 6

- The financial system is a network of intermediaries, markets, instruments, policymakers and regulations.
- Three components, two functions each: create/transfer money; accumulate/deploy savings; market/transfer assets.
- Hierarchy: central bank, depository banks, NBFCs, investment banks, brokerages.
- Three independent axes: maturity (money/capital), newness (primary/secondary), structure (exchange/OTC).
- Eight market categories span debt, equity, derivatives and foreign exchange.

PART C • PARTICIPANTS AND MECHANICS

Who participates, and how the machinery works

7 The Four Types of Investors

PART C • PARTICIPANTS AND MECHANICS • TOPIC 7 OF 9 • SERVES LEARNING OBJECTIVE 3



IN ONE LINE

Four investors can hold the same share at the same price on the same day for four completely different reasons.

Why this matters

Investor type determines instrument preference, holding period and risk appetite — which is why the same news moves different holders in different directions. Week 3 builds directly on this when it matches instruments to holders.

Same share, four reasons

One Infosys share trades on the NSE at, say, ₹4,800. Four very different holders might own it: a twenty-four-year-old in Pune investing ₹5,000 of his first bonus through Groww; a life insurer deploying policyholder money for long-term stability; a sovereign wealth fund diversifying national reserves across borders; and a state pension fund sitting on surplus cash it will not need for twenty years.

What separates them

Individual investors invest their own savings, bear the full benefit and consequence of their choices, and vary widely in risk appetite by age and income.

Institutional investors invest on behalf of others — this is the key differentiator. A pension fund deploys the retirement savings of thousands; an insurer manages premium income. Their scale means a single large trade can move a price, and their instrument preferences follow their liabilities.

Government investors deploy surplus public funds, prioritising capital preservation and liquidity over growth. Sovereign wealth funds are the large-scale example.

Foreign investors cross borders for geographic diversification and higher returns. They bring capital, improve liquidity and aid price discovery — and the same mobility introduces volatility when conditions shift and capital withdraws rapidly.

FOUR HOLDERS • ONE SECURITY • FOUR OBJECTIVES

INDIVIDUAL

WHOSE MONEY

Their own savings

OBJECTIVE

Wealth, education, a purchase, retirement

PREFERS

Stocks, bonds, mutual funds

INSTITUTIONAL

WHOSE MONEY

Other people's, held in trust

OBJECTIVE

Meet obligations to beneficiaries

PREFERS

Instruments matched to liability timing

GOVERNMENT

WHOSE MONEY

Surplus public funds

OBJECTIVE

Capital preservation and liquidity

PREFERS

Treasury and low-volatility instruments

FOREIGN

WHOSE MONEY

Home capital deployed abroad

OBJECTIVE

Diversification and higher returns

PREFERS

Cross-border equity and debt

Figure 7.1 The four investor types. The differentiator for institutions is whose money is at stake.

KEY IDEA

The defining feature of an institutional investor is not size but **stewardship**: it invests other people's money. That single fact explains its instrument choices, its holding period and the regulation around it.

EXAM POINTER

Answer investor-type questions on four axes: whose money, objective, scale, and instrument preference. The mark most often missed is the link between a pension fund's *long-term liabilities* and its preference for long-dated assets. State the causal chain, do not merely assert the preference.

QUICK RECAP • TOPIC 7

- Individual investors use their own savings; risk appetite varies with age and income.
- Institutional investors deploy other people's money — the key differentiator — at market-moving scale.
- Their instrument preference follows the timing of their liabilities.
- Government investors prioritise preservation and liquidity; sovereign wealth funds are the large example.
- Foreign investors add capital and liquidity, and also volatility when they withdraw.

8 The Flow of Funds and Financial Intermediaries

PART C • PARTICIPANTS AND MECHANICS • TOPIC 8 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

Funds move from lender-savers to borrower-spenders by two routes, and intermediaries exist because the direct route is defeated by three specific problems.

Why this matters

In September 2008 the flow of money itself broke. Banks stopped lending to each other, companies could not make payroll, and firms with no connection to Wall Street could not borrow to buy inventory. Understanding the plumbing is what makes that comprehensible.

Two routes

On one side sit **lender-savers**: households, firms, governments and foreign participants with surplus funds. On the other sit **borrower-spenders**, needing funds beyond current income. The financial system sits between.

In **direct finance**, borrowers raise funds straight from lenders through markets by issuing instruments — a company selling bonds to investors. In **indirect finance**, funds route through an intermediary such as a bank. Most financing worldwide happens indirectly, which makes intermediaries more important than securities markets in funding corporations globally — a fact that surprises most students.

The three problems intermediaries solve

Transaction costs. Direct lending involves time, legal fees and effort. Intermediaries reduce these through expertise and economies of scale, spreading them across many transactions, which makes even small loans viable.

Risk. Intermediaries perform *asset transformation* — borrowing short and lending long. A bank accepts short-term deposits and makes long-term loans, converting risky illiquid assets into safer liquid ones for depositors, and profiting from the spread.

Asymmetric information. The borrower knows more than the lender. Before the transaction this produces *adverse selection* — the riskiest borrowers are the keenest to borrow — addressed by screening. After it, *moral hazard* — borrowers take greater risks once funded — addressed by monitoring.

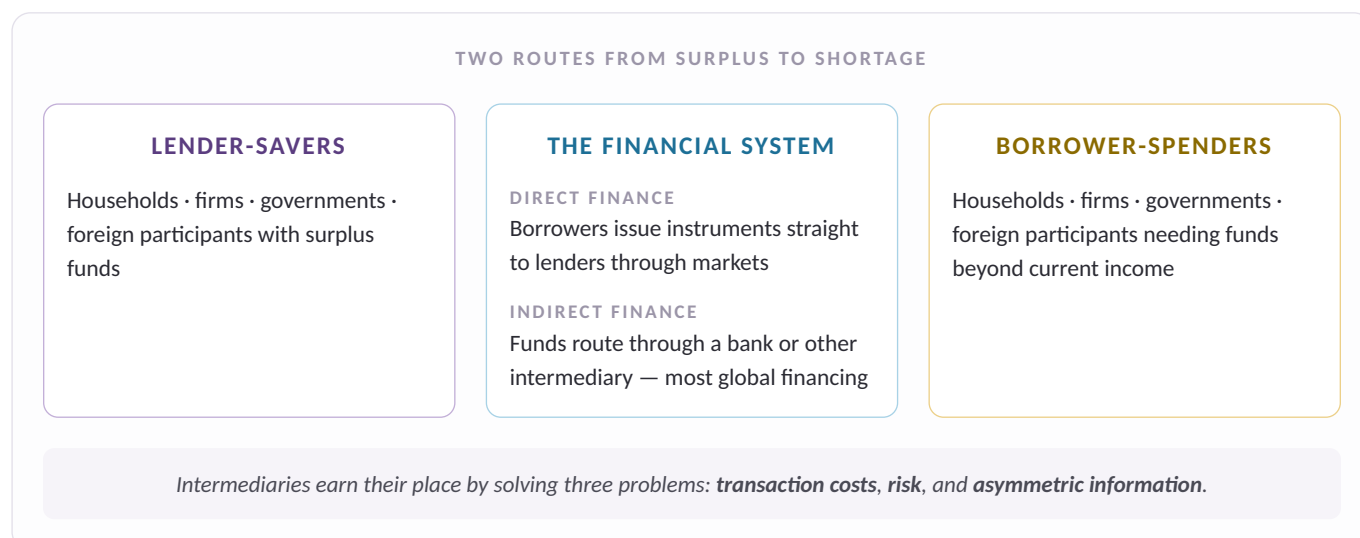


Figure 8.1 The flow of funds, and the two routes it can take.

COMMON PITFALL

Confusing adverse selection with moral hazard. The test is timing. Adverse selection happens *before* the loan and is fixed by screening; moral hazard happens *after* and is fixed by monitoring. Examiners use this pair precisely because the two are so often swapped.

EXAM POINTER

A full answer names both routes, states that indirect finance dominates globally, then gives all three problems *with their solutions*: scale for transaction costs, asset transformation for risk, screening and monitoring for asymmetric information. Finish with the three intermediary categories — depository, contractual savings, investment.

QUICK RECAP • TOPIC 8

- Funds flow from lender-savers to borrower-spenders through the financial system.
- Direct finance issues instruments to lenders; indirect finance routes through intermediaries.
- Most financing worldwide is indirect — intermediaries matter more than securities markets.
- Three problems solved: transaction costs, risk (asset transformation), asymmetric information.
- Adverse selection precedes the loan (screening); moral hazard follows it (monitoring).

9

Price Discovery

PART C • PARTICIPANTS AND MECHANICS • TOPIC 9 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

Price discovery is how markets turn information into prices — and those prices decide how capital is allocated across the whole economy.

Why this matters

This is the closing idea of the week and the one that ties the rest together. Every earlier topic — markets, investors, intermediaries — exists to make this process work.

The mechanism

On 3 March 2020 the Sensex stood at 25,981 and analysts spoke of 20,000, even 15,000. By 21 January 2021 it crossed 50,000 for the first time. Same companies, same factories, same workers. What changed was information, expectations, and the collective judgement of millions of buyers and sellers.

Price discovery is the process by which buyers and sellers interact to determine a security's price. The buyer will pay up to a certain price — the **bid**. The seller will accept no less than a certain price — the **ask**. When the two converge, a trade occurs and a price is established.

Information, and why depth matters

Price discovery does not happen in a vacuum. In an efficient market, prices reflect all available public information; when new information arrives, investors reassess and prices adjust rapidly. A price at any moment is the aggregated judgement of every active participant, not the view of one analyst — which is why deeper, more liquid markets discover prices more accurately.

Why it decides who gets capital

New issues are not priced arbitrarily; they are tied to secondary market prices for comparable securities. If a company's shares trade at ₹500, it cannot price a new issue at ₹700 — the market has already told it what the share is worth, and new stock is typically priced at a small discount to attract buyers. The same holds for bonds: rising secondary yields signal greater perceived risk and raise the cost of the next issue.

So accurate prices route capital to its most productive uses. A firm with strong prospects commands a higher price, raises capital more cheaply, invests and employs. A weak firm faces a lower price and higher borrowing costs. Distorted prices — from manipulation, opacity or thin trading — misallocate capital and reduce economic efficiency, which is why regulators invest so heavily in market integrity.

HOW A PRICE IS DISCOVERED • BID AND ASK CONVERGE



Repeat across thousands of securities every trading day, and the market's collective estimate of value is continuously refined.

Figure 9.1 Price discovery in action. The trade is the moment information becomes a price.

KEY IDEA

Price discovery is not bookkeeping. It is the mechanism that decides which firms can raise money cheaply and which cannot — which makes an accurate secondary market a precondition for efficient investment across the entire economy.

REAL-WORLD EXAMPLE

The Sensex moved from 25,981 to above 50,000 in ten months with no change in the underlying factories or workforces. The revaluation was entirely a revision of expectations, arrived at through millions of individual bid-and-ask decisions.

EXAM POINTER

Define bid, ask and convergence; state that prices reflect aggregated public information; then make the link examiners are actually testing — secondary market prices set the terms on which primary issues can be sold. Close with the consequence: accurate prices allocate capital efficiently, distorted prices do not.

QUICK RECAP • TOPIC 9

- Price discovery is buyer and seller interaction determining a security's price.
- The bid is the highest price a buyer will pay; the ask the lowest a seller will accept.
- Prices reflect aggregated public information and adjust rapidly to new information.
- Secondary market prices set the terms for primary issues — a company cannot price above what the market says.
- Accurate prices route capital to its most productive uses; distorted prices misallocate it.

Module summary, by learning objective

Organised the way assessment will test you, rather than in topic sequence.

LO	What you should now be able to do	Where it was covered
1	Define finance, name the three core decisions, and describe the six components and four nested layers	Topics 1, 2 and 3 • Figures 1.1, 2.1, 3.1
2	Explain the six principles and apply the time-value-of-money formula	Topics 4 and 5 • Figures 4.1, 5.1
3	Classify markets on three axes and distinguish the four investor types	Topics 6 and 7 • Figures 6.1, 7.1
4	Explain the flow of funds, why intermediaries exist, and how price discovery allocates capital	Topics 8 and 9 • Figures 8.1, 9.1

The module in six sentences

Finance is how individuals, institutions, governments and businesses acquire, spend and manage money, and it reduces to three decisions repeated at every scale. Those decisions happen inside a financial environment of six interdependent components, nested so that investments sit at the core and depend on everything outside them. Economics allocates resources while finance allocates the funds that pay for them, which is why ability to pay matters as much as price. Six principles govern every financial decision, from the time value of money through to trust and ethics. Markets are classified on three independent axes, and four types of investor meet in them, each with different money at stake. Funds flow from savers to borrowers directly or through intermediaries that solve transaction costs, risk and asymmetric information — and price discovery turns all of that activity into the prices that decide who gets capital.

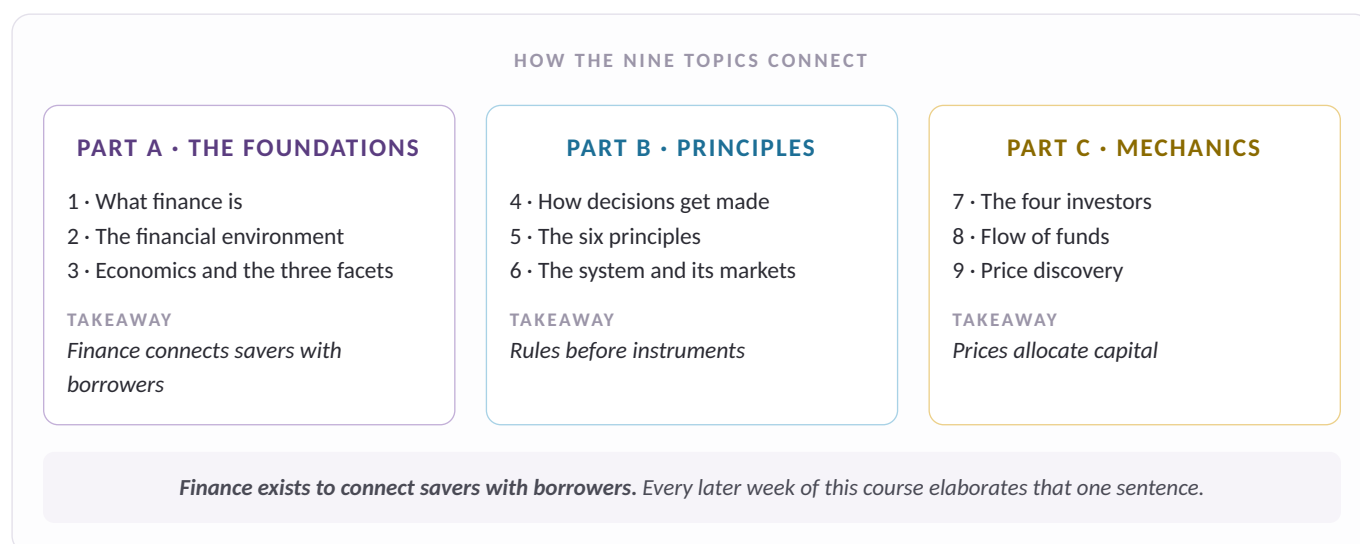


Figure M1 How the nine topics connect. Revise along the parts, not down the list.

Glossary and self-check



Term	Definition
Finance	How individuals, institutions, governments and businesses acquire, spend and manage money and other financial assets.
Financial environment	The six interdependent components within which all financial activity happens.
Nested model	Environment → institutions → markets → investments. Each layer depends on the one outside it.
NBFC	Non-banking finance company. Provides financial services without a full banking licence.
Patient capital	Funding willing to wait for returns over a long horizon, supporting innovation and growth.
Time value of money	A rupee today is worth more than a rupee tomorrow, because it can be invested.
Diversification	Spreading investments across assets to cut risk without proportionally cutting expected return.
Market efficiency	The condition in which prices already reflect all available public information.
Principal-agent problem	The conflict when managers (agents) pursue goals different from shareholders (principals).
Money market	Instruments maturing in one year or less. High liquidity, low risk, modest return.
Capital market	Instruments maturing beyond one year: bonds, mortgages and equity.
Primary market	Where securities are issued for the first time; proceeds go to the issuer.
Secondary market	Where existing securities trade between investors; supplies liquidity and price signals.
Institutional investor	An entity investing other people's money — the defining feature, not size.
Direct finance	Borrowers raise funds straight from lenders by issuing instruments through markets.
Indirect finance	Funds routed to borrowers through an intermediary. Most global financing works this way.
Asset transformation	Borrowing short and lending long, converting illiquid risky assets into safer liquid ones.
Adverse selection	Before the loan: the riskiest borrowers are keenest to borrow. Addressed by screening.
Moral hazard	After the loan: borrowers take greater risks once funded. Addressed by monitoring.
Price discovery	The process by which buyer and seller interaction determines a security's price.

Self-check

Ungraded retrieval practice. Answer from memory first, then check.

Q1 A company's shares trade at ₹800 in the secondary market. Why can it not price a new issue at ₹1,000?

Because the secondary market has already priced the share. A new issue is tied to observed prices for comparable securities, and is typically set at a small discount to attract buyers. The secondary price is the signal the company, its bank and its investors all read.

Q2 Distinguish adverse selection from moral hazard, and give the mechanism that addresses each.

Timing separates them. Adverse selection occurs *before* the transaction — riskier borrowers are the keenest — and is addressed by screening. Moral hazard occurs *after* — borrowers take greater risk once funded — and is addressed by monitoring.

Q3 Calculate the future value of ₹2,00,000 invested at 7% per annum for three years.

$FV = PV \times (1 + r)^t = 2,00,000 \times (1.07)^3 = ₹2,45,009$. Show the formula and the substitution; the method carries the marks, not the figure.

Q4 Two investors hold the same Infosys share. One is a retail investor, one a pension fund. What differs?

Whose money, and therefore the horizon. The retail investor risks their own savings and may hold for personal goals. The pension fund invests other people's money against long-dated liabilities, so it prefers long-term assets and trades at a scale that can move the price.

Q5 Name the three independent ways a financial market can be classified, and label one security on all three.

Maturity, newness and structure. A newly issued government bond bought on an exchange is capital market, primary and exchange-traded simultaneously. The three axes are orthogonal, not a sequence.

Q6 Why is finance not simply a branch of economics?

Economics allocates resources; finance allocates the funds that pay for them. In economics an equilibrium price clears the market. In finance that price must also be supported by the ability to pay — which is what every instrument in the course exists to arrange.
